

NORMAN BUI

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EDUCATION

- Carleton University** | *Bachelor of Computer Science (Honours), 3.9/4.0 GPA* Graduated Spring 2026
- Coursework: Statistical Modeling, Data Structures & Algorithms, Database Management Systems, Linear Algebra
 - Awards: Dean's Honour List every year; President's, Henry Marshall Tory, and Harry S. Southam Scholarships

EXPERIENCE

- Canadian Tire Corporation** | *Software Developer* July 2026 - Present
- Built an internal tool that turns a service's source code into written documentation
 - Grounded every page in facts pulled from the code, so it cannot describe features that do not exist
 - Writes for **3** audiences off the same facts, including a business page whose rules stop the model guessing a company name or an abbreviation
 - Cut the cost of each run **38x** by testing candidate AI models on real projects, with no loss in accuracy
- Electronic Arts (EA)** | *Software Developer Intern* May 2025 - Aug 2025
- Raised automated performance test pass rates from **35% to 100%** by fixing broken logins and setup steps
 - Evaluated an AI system that plays the game to find bugs, measuring how reliably it worked
 - Catalogued its failure patterns and wrote up findings the team used to improve it
- Autodesk** | *Software Developer Intern* Jan 2025 - Apr 2025
- Built systems that find and fix missing manufacturing data in Autodesk Fusion, with a repair step **8x** faster
 - Designed predictable, rule-based checks for repairing manufacturing data
 - Shipped services that check and repair customer data automatically, with **100%** test coverage
- Public Health Agency of Canada** | *Data Scientist Intern* Sep 2024 - Dec 2024
- Built an AI-assisted research review process for epidemiologists, with Imperial College London and the WHO
 - Tested which AI models found the most relevant research, setting the baseline the team tuned against
 - Prototyped automatic collection of academic papers and their key details, replacing manual gathering
- Bedarra Corporation** | *Student Researcher* Aug 2024 - Oct 2024
- Built an AI assistant that answers questions about rarely documented programming languages
 - Compared retrieval approaches to find which returned the most useful answers when documentation is scarce
 - Wired the assistant to pull real code examples and documentation into every answer
- Health Canada** | *Data Scientist Intern* Sep 2023 - Dec 2023
- Built models that predict how toxic a chemical is likely to be, supporting risk assessments
 - Built a pipeline that maps the relationships between chemicals, diseases, and published research
 - Rebuilt the system that gathers chemical data, more than doubling usable sources with **100%** test coverage

PROJECTS

- Orbital Refuelling Simulator** | *Python, Streamlit, scikit-learn, Qwen LoRA, pytest* github.com/nbui23/orbital-refuelling
- Simulated a **410-second** spacecraft refuelling mission across **9** phases, with **9** kinds of failure to catch
 - Combined fixed safety limits with machine learning that flags subtler drift, keeping the two clearly separate
 - Compared rule-based alerts against machine learning scores across every failure scenario
 - Fine-tuned a small AI model to explain each alert in plain language, scoring **4.86/5** on held-out tests
- TwinQuery** | *Python, LangGraph, FastAPI, Streamlit, PostgreSQL/PostGIS, Ollama, RAG, Docker* github.com/nbui23/twinquery
- Built an assistant that lets non-technical users ask plain-English questions about building energy data
 - Turned those questions into safe database queries over Ottawa building data, with guardrails on every one
 - Grounded every answer in both the database and retrofit guidance documents, showing the sources behind it

SKILLS

Languages: Python, TypeScript, JavaScript, C++, Java, SQL
Frameworks & Runtimes: FastAPI, Flask, Streamlit, LangGraph, LangChain, Next.js, Node.js, Spring Boot
AI / ML / Data: LLMs, RAG, LangGraph, LangChain, Ollama, Qwen LoRA, scikit-learn, pandas, NumPy, PyTorch, FAISS
Databases & APIs: PostgreSQL/PostGIS, pgvector, MySQL, SQLite, MongoDB, Firebase, Neo4j, GraphQL
Tools & Practices: Git, Docker, Docker Compose, AWS, CI/CD, Ubuntu, TDD, pytest, unittest, Postman, Agile, SDLC
Spoken Languages: English, French, Vietnamese